Homework

Consider production of rectangular tables with length $\ell = 160\,\text{cm}$ and width $w = 90\,\text{cm}$.

Tolerance of production process (standard deviation) is $\sigma_\ell = 0.5\,\text{cm}$ for the table length and $\sigma_w = 0.3\,\text{cm}$ for the table width. Assuming length and width are independent random variables and table sides are always perpendicular:

1. Calculate (from analytical formulae):
   - the expected variation (standard deviation) of the table surface,
   - the expected variation of the total length of the table sides,
   - the expected variation of the table diagonal
   - the expected correlation between the surface and the total edge length.

2. Perform numerical experiment generating table lengths and widths from the Gaussian distribution to calculate same parameters, compare results of the two approaches.

Solutions should be uploaded until November 11.